

Stage 3 — Real artifacts across modalities

Where the Shortcut Sits: supervisor meeting 3 of 6

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In one sentence. With real, unedited artifacts — hair in dermoscopy, calipers in thyroid and ovarian ultrasound, debris in capsule endoscopy — ROI masking helps far more when the artifact lies outside the ROI than when it lies inside it (crossover positive in 6 of 7 cohort–encoder pairs and in all four pre-specified primary tests after Holm correction); the dermatology encoder DermLIP and real chest-radiograph devices are the exceptions.

Notation

$Y \in \{0, 1\}$	diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).
$A \in \{0, 1\}$	artifact present (hair, sonographer caliper, capsule debris, chest tube).
ROI	region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in green; artifacts in red.
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).
Environments	training and correlated test $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real traps: half of each class; sweeps: absent).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images).
CI	95% crossed seed×image bootstrap interval (Stage 1, Sec. ??); every interval in these reports uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.
SMD	standardised mean difference between two groups.

1 Where we are

Stage 2 established the location law with synthetic artifacts whose position was controlled. Synthetic artifacts answer *whether* location can matter; they cannot show that it matters for the artifacts clinicians actually produce. This meeting moves to real artifacts, whose location is read from masks rather than set by us. It also reports a design that failed (the E12 overlap contrast), how well the automatic artifact masks agree with the images, where the law does not hold (chest radiography), and the pre-specified 16-test confirmatory family. All intervals are from regenerated runs with the corrected bootstrap; the chest-radiograph device runs could not be regenerated and are shown as point estimates only.

2 Question

Q3a With real artifacts and the same protocol, is ROI masking less useful — or harmful — when the artifact lies inside the ROI than when it lies outside it (a positive *crossover*)?

Q3b Does this hold across modalities, artifacts and encoders?

Q3c Are the artifact masks that define “inside” and “outside” good enough to support the answer?

Design: two traps sharing one artifact-free group A real artifact’s location is fixed by the image, so we cannot move it. Instead each cohort is split into images whose artifact lies mostly inside the ROI (overlap $r \geq 0.5$, *Trap A*) and images whose artifact lies outside it ($r < 0.1$, *Trap B*). Each trap is completed by the same artifact-free images and receives the trap environments of Stage 1 (training and correlated test 0.9/0.1, reversed test 0.1/0.9). The primary quantity is the *crossover* $[\text{mask} - \text{ERM}]_B - [\text{mask} - \text{ERM}]_A$ on the reversed test: the law predicts it is positive.

Rejected: the overlap-contrast design (E12) The pilot compared inside-ROI with outside-ROI artifacts directly, without an artifact-free group. That design makes “artifact inside” versus “artifact outside” the label proxy; masking then removes exactly the outside artifacts that define one side of the contrast and keeps the inside ones that define the other, so masking is expected to harm by construction and the result says nothing about location. Section 5.4 shows it.

3 Methods

Cohorts Table 2. Dermoscopy: ISIC 2019 (Combalia et al., 2019; Tschandl et al., 2018) with the published semi-automatic hair masks (Wegley et al., 2026); lesion masks are HAM10000 manual masks or, for BCN and MSK images, a U-Net trained on ISIC 2018 Task 1 only (Codella et al., 2019). Thyroid: TN3K images (Gong et al., 2021) with the TNCD malignancy labels and nodule masks; calipers segmented by a rule-based detector. Ovary: MMOTU (Zhao et al., 2022) tumour masks and a caliper detector. Capsule endoscopy: SEE-AI (Yokote et al., 2023) expert lesion boxes and debris masks from a patch-token probe trained on expert-masked frames.

Protocol (E13) Frozen encoders (DINOv2 ViT-B/14 @518 for every cohort; DermLIP for dermoscopy, MedSigLIP-448 for thyroid and capsule), logistic heads, 5 seeds $\times$ 5 grouped folds (groups: lesion identifier and perceptual hash; capsule: blocks of 100 frame numbers plus hash duplicates, since no video identifiers are published). Within each label the artifact-bearing images are subsampled to the source mix of the artifact-free group (source matching).

Statistics Crossed seed $\times$ image bootstrap, 10,000 replicates; the crossover’s interval resamples the same images in both traps. The 16-test primary family is Holm-corrected; everything else is reported with unadjusted 95% intervals.

4 Experiments

Table 1: Experiments of this stage with registration status, first commit of the registration and the support criterion stated in advance. Commit times come from this machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local)	Support criterion
Dermoscopy hair traps (E13, DINOv2)	supported registered	ISIC2019_ TRAPS	76e25d4, 13:14	2026-09-26 crossover CI > 0
Dermoscopy hair, DermLIP	supported registered	ISIC2019_ TRAPS	76e25d4, 13:14	2026-09-26 crossover CI > 0
Thyroid caliper traps	supported registered	THYROID_ TRAPS	ea57661, 22:46	2026-09-26 crossover CI > 0
Ovarian caliper traps	supported registered	OVARY_TRAPS	2517600, 07:29	2026-09-27 crossover CI > 0
Capsule debris traps	supported registered	CAPSULE_ TRAPS	fcfbf8d9, 00:40	2026-09-27 crossover CI > 0
Chest-radiograph device traps	supported registered	CXR_DEVICE_ TRAPS	4d1a924, 18:07	2026-09-26 crossover CI > 0
E12 overlap-contrast trap	partly supported pilot design, replicated	REPLICATION_ SPEC	2699d51, 15:47	2026-09-26 null (pilot result)
16-test primary family (Holm)	not supported grouped after the individual results	STATISTICAL_ PLAN	4e9d44c, 09:37	2026-09-27 Holm $p < 0.05$, positive estimate
Regeneration with the crossed bootstrap	supported registered	FINAL (A1)	c45d482, 00:35	2026-09-28 claims whose CI now includes 0 are rewritten

Caveat on the primary family. The 16 tests were grouped into one Holm family after the individual results were known (the statistical plan 4e9d44c, 2026-09-27 09:37 was written after the first trap results). Each test was registered with its cohort; the grouping is therefore conservative bookkeeping, not a pre-registered family.

5 Results

5.1 Cohorts and masks

Table 2: Real-artifact cohorts and pool sizes (images, positive/negative diagnosis) before the environments are sampled. Trap A: artifact mostly inside the ROI ($r \geq 0.5$); Trap B: outside ($r < 0.1$); both share the artifact-free group.

Cohort	Task	Artifact source)	(mask	ROI	Artifact-free pos/neg	Trap A pos/neg	Trap B pos/neg
ISIC 2019 dermoscopy (HAM10000, BCN20000, MSK)	melanoma vs other	hair masks)	(published	lesion: HAM manual / U-Net	157/706	942/2655	969/4412
TN3K ultrasound, TNCD labels	malignant vs benign nodule	calipers (rule de- tector)	(rule de-	nodule (manual)	749/1052	292/858	99/227
MMOTU 2D ultrasound	malignant vs benign tumour	calipers (rule de- tector)	(rule de-	tumour (man- ual)	178/170	276/341	103/91
SEE-AI dermoscopy	erosion vs polyp-like lesion	debris probe)	(patch	lesion box (ex- pert)	343/164	412/404	1009/231

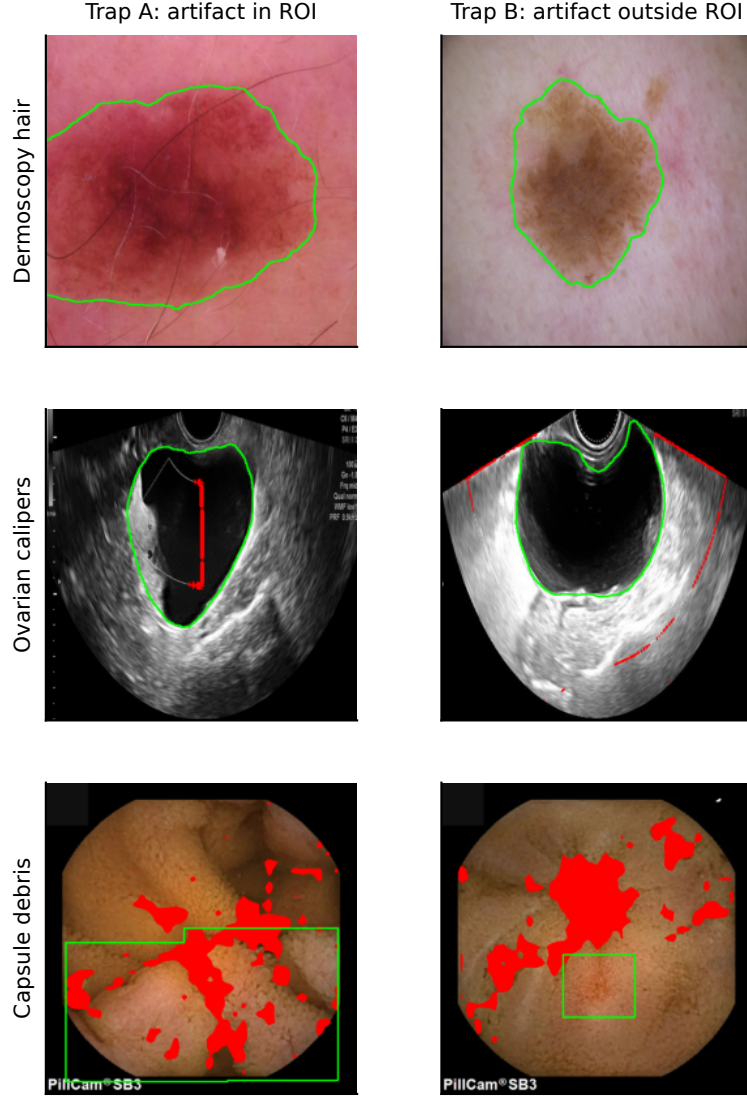


Figure 1: Real artifacts inside (Trap A) and outside (Trap B) the ROI. **Green**: ROI; **red**: automatic artifact mask. Dermoscopy is shown with the lesion outline only, because the published hair masks carry no licence that allows redistribution (overlays are kept locally). Thyroid images are not shown (data licence not stated). Automatic overlap of the shown images: dermoscopy 0.534 and 0.000; ovary 1.000 and 0.000; capsule 0.740 and 0.005. ISIC CC BY-NC 4.0; MMOTU, SEE-AI CC BY 4.0.

Mask validity Table 3. Lesion masks are accurate (U-Net held-out Dice 0.886; against HAM manual masks 0.895). The capsule debris probe reaches pixel accuracy 0.922 and IoU 0.625 against expert masks. The thyroid caliper detector’s detections lie on a real caliper in 38 of 40 audited images. The ovarian “marker-free” group is caliper-free but not overlay-free. The morphological hair detector used for the pilot’s counterfactual (E10) is poor (IoU 0.222), which is why all hair traps use the published masks. Label noise in the artifact masks dilutes the constructed correlation and biases effects toward zero; it cannot create a location difference by itself.

Table 3: Validity of the automatic masks that define artifact presence and location. Visual audits: docs/ARTIFACT_MASK_AUDIT.md (by the analysis agent, not a clinician; Stage 6 reports the blinded image audit prepared for a clinician).

Mask	Source	Check	Result
ISIC 2019 lesion masks (BCN, MSK)	U-Net trained on ISIC 2018 Task 1	held-out Dice (mean) / Dice vs HAM manual masks	0.886 / 0.895
Hair (DullRazor-type detector, E10)	morphological	IoU / precision / recall vs Mendeley masks	0.222 / 0.316 / 0.472
Capsule debris masks	patch-token probe (2,160 expert-masked frames)	held-out pixel accuracy / IoU	0.922 / 0.625
Thyroid (TN3K), detected (A1)	visual audit, 40 images	detection lies on a real caliper / measurement mark	38/40 (95 %); 2 border false positives; 1 dotted zoom box (overlay)
Thyroid, marker-free (A0)	visual audit, 40 images	no visible caliper	38–39/40 (95–97 %); depth-scale ticks present in all images (not label-related)
Ovary (MMOTU), detected (A1)	visual audit, 40 images	an on-screen overlay is present	40/40; caliper specifically 31/40 (others: dotted zoom arcs, focus markers, thumbnail inset)
Ovary, marker-free (A0)	visual audit, 40 images	no visible caliper	37/40 (92 %)
Ovary, marker-free (A0)	visual audit, 40 images	no overlay of any kind (arrows, boxes)	~29/40 (72 %); text labels common

5.2 The crossover with real artifacts

Table 4 and Figure 2. With DINOv2, masking is worse for in-ROI artifacts in every cohort:

- **Dermoscopy hair.** Masking *harms* when the hair lies on the lesion ($-0.049 [-0.083, -0.018]$) and helps when it lies outside ($+0.098 [+0.060, +0.136]$); crossover $+0.147 [+0.111, +0.185]$. After masking, the Trap A model still separates the correlated test (0.917) far better than the reversed one (0.460): the retained hair is used. The harm comes mainly from BCN images ($-0.067 [-0.110, -0.025]$; HAM $-0.011 [-0.064, +0.043]$; Table 5; MSK is not reported separately).
- **Thyroid calipers.** Masking helps in both traps but much more when the calipers lie outside the nodule ($+0.348 [+0.307, +0.388]$ versus $+0.109 [+0.085, +0.133]$; crossover $+0.239 [+0.196, +0.280]$); with MedSigLIP the crossover is $+0.195 [+0.153, +0.238]$.
- **Ovarian calipers.** Crossover $+0.170 [+0.087, +0.252]$ (Trap A $+0.132 [+0.072, +0.196]$, Trap B $+0.302 [+0.225, +0.378]$).
- **Capsule debris.** The largest crossover ($+0.377 [+0.327, +0.426]$; MedSigLIP $+0.335 [+0.286, +0.383]$): masking recovers almost all of the reversed-test loss when the debris lies outside the lesion box and little when it lies on it.

The law in its real-artifact form is therefore a *crossover*, and whether masking actually harms in Trap A depends on the cohort (it does for dermoscopy hair; for calipers and debris it helps less).

Table 4: Real-artifact traps: change in reversed-test AUROC from ROI masking, per trap, and the crossover (5 seeds $\times$ 5 grouped folds; crossed 95% CIs).

Cohort	Encoder	ERM reversed AUROC A / B	mask – ERM, Trap A (in ROI)	mask – ERM, Trap B (outside)	Crossover B – A
Dermoscopy hair	DINOv2	0.510 / 0.510	$-0.049 [-0.083, -0.018]$	$+0.098 [+0.060, +0.136]$	$+0.147 [+0.111, +0.185]$
Dermoscopy hair	DermLIP	0.665 / 0.735	$-0.104 [-0.143, -0.067]$	$-0.090 [-0.135, -0.046]$	$+0.014 [-0.037, +0.063]$
Thyroid calipers	DINOv2	0.289 / 0.405	$+0.109 [+0.085, +0.133]$	$+0.348 [+0.307, +0.388]$	$+0.239 [+0.196, +0.280]$
Thyroid calipers	MedSigLIP	0.289 / 0.481	$+0.053 [+0.032, +0.073]$	$+0.248 [+0.200, +0.293]$	$+0.195 [+0.153, +0.238]$
Ovarian calipers	DINOv2	0.413 / 0.423	$+0.132 [+0.072, +0.196]$	$+0.302 [+0.225, +0.378]$	$+0.170 [+0.087, +0.252]$
Capsule debris	DINOv2	0.581 / 0.429	$+0.084 [+0.055, +0.112]$	$+0.461 [+0.415, +0.506]$	$+0.377 [+0.327, +0.426]$
Capsule debris	MedSigLIP	0.570 / 0.464	$+0.117 [+0.088, +0.147]$	$+0.452 [+0.407, +0.495]$	$+0.335 [+0.286, +0.383]$

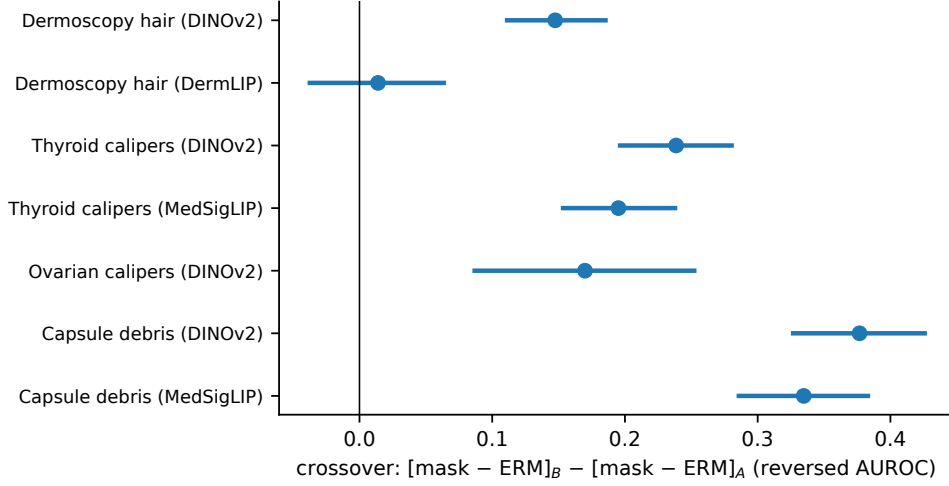


Figure 2: Crossover with real artifacts, per cohort and encoder (crossed 95% CIs).

Table 5: Dermoscopy hair, Trap A, by image source (DINOv2).

Source	mask - ERM, Trap A
HAM	-0.011 [-0.064, +0.043]
BCN	-0.067 [-0.110, -0.025]

Hair area relative to the lesion In Trap A the in-lesion hair covers a median 0.033 of the lesion’s area — similar to the synthetic ruler of Stage 2. Split into tertiles of that ratio, masking harms in every tertile (Table 6); unlike the synthetic sweep, no gradient across tertiles is visible, and the intervals overlap.

Table 6: Dermoscopy hair, Trap A, by the ratio of in-lesion hair area to lesion area.

Hair/lesion area tertile	Median ratio	mask - ERM, Trap A
tertile 1	0.006	-0.057 [-0.095, -0.022]
tertile 2	0.033	-0.046 [-0.082, -0.013]
tertile 3	0.113	-0.037 [-0.071, -0.005]

5.3 The exception among encoders: DermLIP

With DermLIP masking harms at *both* hair locations (Trap A -0.104 [-0.143, -0.067], Trap B -0.090 [-0.135, -0.046]) and the crossover vanishes (+0.014 [-0.037, +0.063]). DermLIP was trained on whole dermatology images and loses disease signal when the background is removed, whatever the hair does. Stage 5 analyses this exception with the theory.

5.4 The failed design: E12 overlap contrast

Table 7: E12 overlap-contrast design: hair inside the lesion ($A=1$, $r \geq 0.5$) versus hair outside it ($A=0$, $r < 0.1$), no artifact-free group (DINOv2, 5 seeds $\times$ 5 folds).

Arm	change vs ERM (reversed AUROC)
ROI masking	-0.103 $[-0.121, -0.085]$
group-balanced	$+0.160$ $[+0.144, +0.177]$
DFR	$+0.179$ $[+0.148, +0.211]$

The pilot reported E12 as null. The re-implementation finds masking clearly harmful (-0.103 $[-0.121, -0.085]$), while group balancing ($+0.160$ $[+0.144, +0.177]$) and DFR ($+0.179$ $[+0.148, +0.211]$) help. As argued above, this is what the design predicts whatever the location effect, because the label proxy *is* location. The design was abandoned in favour of the two-trap design with a shared artifact-free group; the pilot’s null result is not supported.

5.5 Chest radiography: the boundary

In NIH radiographs linked to RANZCR-CLiP device annotations, Trap A contains central venous catheters inside the lungs and Trap B endotracheal tubes outside them. Lung masking helped for both device types in all four registered findings (Table 8); the registered claim that masking harms (or helps less) for the in-lung catheter was not supported. The registration itself notes a confound: the catheter-free group carries many more out-of-lung tubes, which masking removes. Together with the synthetic-tube sweep of Stage 2, where RAD-DINO ignored out-of-lung tubes, chest radiography is the boundary of the law. The intervals of these runs were produced with the old estimator and cannot be regenerated (the per-image predictions were not saved and the RANZCR data are not on this machine); they are withdrawn and only point estimates are shown.

Table 8: Chest radiographs with real devices (NIH ChestX-ray14 linked to RANZCR-CLiP; RAD-DINO): Trap A = central venous catheter inside the lungs, Trap B = endotracheal tube outside them. Point estimates only, **not re-estimated** with the corrected bootstrap (per-image predictions were not saved; the old intervals are withdrawn).

Finding (NIH, RANZCR-CLiP devices)	mask – ERM, Trap A	Trap B	Crossover
Atelectasis	$+0.045$	$+0.105$	$+0.060$
Consolidation	$+0.070$	$+0.068$	-0.002
Effusion	$+0.104$	$+0.100$	-0.004
Infiltration	$+0.094$	$+0.063$	-0.032

5.6 The 16-test primary family

Table 9. After Holm correction over all 16 tests, 11 of 16 are supported; all 4 location-crossover tests (P1) are supported. P2–P4 test remedies and are discussed in Stage 6. Compared with the old estimator, one test lost support (P3, ovary: Holm p 0.1304).

Table 9: The pre-specified 16-test primary family (Trap A, reversed test unless stated), crossed bootstrap, two-sided p , Holm over all 16 tests. P1 is this stage’s claim; P2–P4 concern the remedies of Stage 6.

Family	Cohort	Estimate [95% CI]	p	p (Holm, 16)	Supported
P1 location crossover	ISIC hair (DINOv2)	+0.147 [+0.111, +0.185]	0.0001	0.0016	supported
P1 location crossover	Thyroid (DINOv2)	+0.239 [+0.196, +0.280]	0.0001	0.0016	supported
P1 location crossover	Capsule (DINOv2)	+0.377 [+0.327, +0.426]	0.0001	0.0016	supported
P1 location crossover	Ovary (DINOv2)	+0.170 [+0.087, +0.252]	0.0001	0.0016	supported
P2 U-MtE (protected) > mask	ISIC hair	+0.042 [+0.024, +0.058]	0.0001	0.0016	supported
P2 U-MtE (protected) > mask	Thyroid	+0.283 [+0.237, +0.329]	0.0001	0.0016	supported
P2 U-MtE (protected) > mask	Capsule	−0.006 [−0.018, +0.006]	0.2946	0.3486	not supported
P2 U-MtE (protected) > mask	Ovary	+0.043 [+0.023, +0.065]	0.0001	0.0016	supported
P3 erase > augment	ISIC hair	+0.035 [+0.017, +0.055]	0.0002	0.0016	supported
P3 erase > augment	Thyroid	+0.215 [+0.189, +0.244]	0.0001	0.0016	supported
P3 erase > augment	Capsule	−0.139 [−0.158, −0.120]	0.0001	0.0016	not supported (opposite)
P3 erase > augment	Ovary	+0.035 [+0.003, +0.067]	0.0326	0.1304	not supported
P4 U-MtE+balanced > balanced	ISIC hair	−0.028 [−0.064, +0.007]	0.1162	0.3486	not supported
P4 U-MtE+balanced > balanced	Thyroid	+0.110 [+0.079, +0.142]	0.0001	0.0016	supported
P4 U-MtE+balanced > balanced	Capsule	+0.079 [+0.042, +0.115]	0.0002	0.0016	supported
P4 U-MtE+balanced > balanced	Ovary	+0.031 [−0.014, +0.077]	0.1730	0.3486	not supported

6 What failed or changed

- **E12 (pilot null) is not supported** and the design was abandoned for a structural reason.
- **Chest-radiograph device traps:** the registered in-ROI harm was not supported (0 of 4 findings); their old intervals are withdrawn because they could not be regenerated.
- **DermLIP has no hair crossover;** the claim “the crossover holds for every encoder” is withdrawn.
- **Primary family:** one test lost support with the corrected bootstrap (P3, ovary); 11 of 16 remain.
- **The source strata of the hair trap** show that the Trap A harm is carried by BCN images.
- **The artifact masks were audited by the analysis agent, not by a clinician;** the blinded clinician audit is prepared (Stage 6) but not yet done.

7 Conclusion

Supported. With real artifacts, masking is less useful when the artifact lies inside the ROI: the crossover is positive in all four cohorts with DINOv2 and in both cohorts tested with MedSigLIP, and all four pre-specified crossover tests survive Holm correction over 16. **Qualified.** Only for dermoscopy hair does masking actually harm in the in-ROI trap; for calipers and debris it helps less. **Not supported.** A crossover with the dermatology encoder DermLIP; an in-ROI penalty for real chest-radiograph devices; the pilot’s null E12 result.

8 Questions for the supervisor

1. Is it acceptable that the artifact masks were validated by visual audit of the analysis agent until the clinician audit (Stage 6) is done, or must the clinician audit precede submission?
2. The 16-test family was grouped after the individual results. Should the thesis present it as a post hoc multiplicity safeguard (as here) or drop it in favour of per-cohort registration only?
3. Should chest radiography stay in the paper as a boundary case with point estimates only, or be moved to the supplement until the device runs can be regenerated?

9 Next stage

Real artifacts are not placed at random: the images whose hair lies on the lesion differ from those whose hair lies beside it. Stage 4 asks whether the crossover survives when those differences are removed — by matching on measured covariates, by regression adjustment on the matched sample, and by moving the *same* real artifact instance inside and outside the ROI of the *same* image — and it reports the paste-edge control and every claim the bootstrap correction changed.

References

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